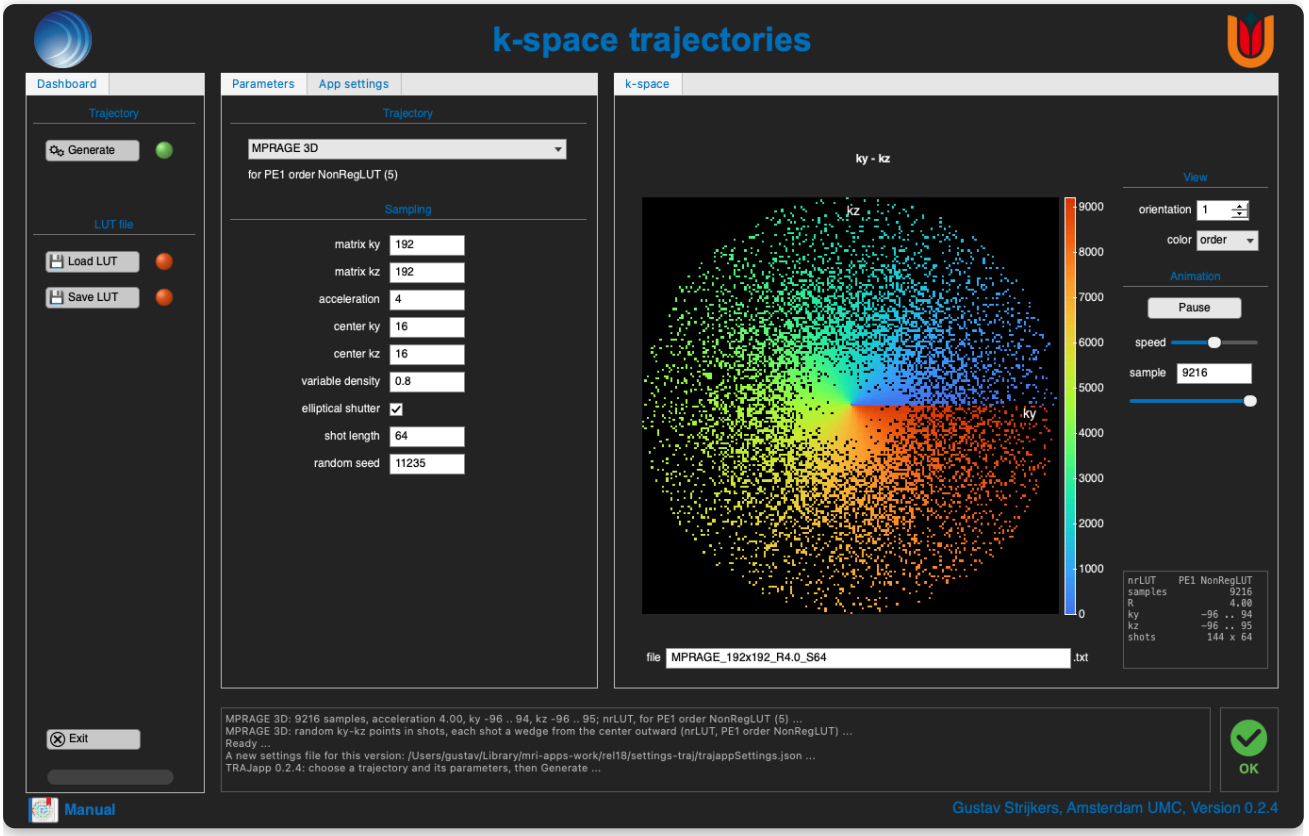


Table of contents

1. Overview
2. Contact
3. Disclaimer
4. Installation
5. Running the software
6. The trajectories
7. Basic operation
8. Using a LUT file on the scanner
9. Reconstruction
10. The settings
11. Credits

1. Overview

TRAJapp is an application with a graphical user interface that makes k-space trajectories for the MR Solutions preclinical MRI systems: the order in which a scan acquires its phase-encoding steps (ky, and kz in 3D), with as many or as few of them as the trajectory asks. It makes undersampled patterns (GRAPPA, CAIPIRINHA, Gaussian, Poisson, MPRAGE shots) and radial and spiral orders, shows how they fill k-space, sample by sample, and saves them as look-up table files (LUT files) that the scanner's FLASH sequence reads.

TRAJapp is the companion of P2ROUD, which reconstructs the scans made with these files.

TRAJapp is a stand-alone application: everything it uses is inside it, and nothing else has to be installed.

2. Contact

TRAJapp and this manual are written by:

Gustav Strijkers
Amsterdam UMC
Biomedical Engineering and Physics
Amsterdam, the Netherlands
g.j.strijkers@amsterdamumc.nl

3. Disclaimer

This is a beta version, the first of a new application. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. Try a new LUT file on a phantom before you use it in a study. In case of questions or issues, please contact Gustav Strijkers.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

TRAJapp comes as a disk image, `TRAJapp-0.2.9-macOS-arm64.dmg`. Nothing else has to be installed.

1. Open the disk image and drag **TRAJapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying TRAJapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/TRAJapp.app
   ```

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

TRAJapp comes as an installer, `TRAJapp-0.2.9-setup.exe`. Run it; when Windows stops it ("Windows protected your PC"), click **More info** and then **Run anyway**. To remove the app, use **Uninstall TRAJapp** in the Start menu, or **Settings > Apps**.

5. Running the software

Start TRAJapp as any other application. It opens with the trajectory and parameters of the last time, k-space still empty: nothing is made until **Generate** is pressed. The message box says what each step did; the newest message is at the top. The **STATUS** picture shows whether the app is busy, ready, or has a warning or an error. The **Manual** button at the bottom left opens this manual. A LUT file given on the app's command line, as a program that starts the app with a file does, is loaded at once, as **Load LUT** would load it. When the app is open already, a file given this way goes to the open window instead: the window comes to the front and loads it, or loads it when the step under way ends.

6. The trajectories

The **Trajectory** drop-down (Parameters tab) offers eight trajectories. Each makes a LUT file of one of two kinds, for one of the scanner's **PE1 order** choices (section 8): an **nrLUT** for **NonRegLUT**, with fewer samples than the matrix has points (undersampled), or an **exLUT** for **ExLUT**, with as many. The line under the drop-down names the PE1 order of the trajectory chosen.

Trajectory	LUT file	PE1 order	What it makes
Radial 3D	exLUT	ExLUT	ky-kz spokes through the center, at tiny golden angles
Spiral 3D	exLUT	ExLUT	ky-kz spirals from the center, at tiny golden angles
Poisson 3D	nrLUT	NonRegLUT	random ky-kz points, denser toward the center, with a fully sampled center
GRAPPA 3D	nrLUT	NonRegLUT	a regular ky-kz grid (every Ry-th ky, every Rz-th kz) with a fully sampled center (ACS)
CAIPIRINHA 3D	nrLUT	NonRegLUT	a ky-kz grid as GRAPPA 3D's, its ky points shifted by one from one sampled kz to the next, with a fully sampled center (ACS)
MPRAGE 3D	nrLUT	NonRegLUT	random ky-kz points in shots, each shot a wedge from the center outward
Gaussian 2D	nrLUT	NonRegLUT	random ky lines, denser toward the center, with fully sampled center lines
GRAPPA 2D	nrLUT	NonRegLUT	every R-th ky line, with fully sampled center lines (ACS)

Their parameters (the **Sampling** panel shows those of the trajectory chosen):

Parameter	Trajectories	What it sets
matrix ky, **matrix kz**	all (kz: 3D)	the number of ky and kz points: the scan's Views and Views 3D
acceleration	Poisson, MPRAGE, Gaussian, GRAPPA 2D	the matrix over the samples acquired
acceleration ky, **acceleration kz**	GRAPPA 3D, CAIPIRINHA 3D	the steps of the grid along ky and kz (whole numbers)
center ky, **center kz** (ACS, center lines)	all but radial and spiral	the fully sampled center, in points
Gaussian width	Gaussian 2D	the width of the lines' density, as a fraction of the matrix
trials	Gaussian 2D	the number of random patterns tried: the one with the cleanest point spread function is kept
variable density	Poisson, MPRAGE	how much denser the samples are toward the center (0: uniform; 0.8 typical)
elliptical shutter	Poisson, MPRAGE	only the points within the ellipse that touches the matrix' edges
shot length	MPRAGE	the samples per shot (a whole divisor of matrix kz)
golden angle	Radial, Spiral	the tiny golden angle between successive spokes or spirals
alternating	Radial, Spiral	every second spoke or spiral runs the other way
random seed	Poisson, MPRAGE, Gaussian, Spiral	the same seed makes the same pattern again

Each trajectory keeps its own values, also from one run of the app to the next.

7. Basic operation

The **Dashboard** at the left holds the steps: **Trajectory** (**Generate**) and **LUT file** (**Load LUT**, **Save LUT**). The lamp beside each is green when it is done, yellow while it runs, and red when it is not done, or no longer matches the parameters.

Step 1: Choosing a trajectory

Choose the trajectory in the **Trajectory** drop-down of the Parameters tab, and set its parameters in the **Sampling** panel below it (section 6). The line under the drop-down names the PE1 order its file is for (for PE1 order NonRegLUT (5) or ExLUT (4); section 8); the message box says what the trajectory makes.

Step 2: Making it

Press **Generate**. The trajectory is made with the parameters set and shown filling k-space (Step 3); the message box gives its samples, its acceleration and its ky and kz ranges, and the kind of LUT file it makes. The lamp stays green while the trajectory shown matches the parameters; a parameter changed turns it red until the next **Generate**, and **Save LUT** waits for that.

A parameter the trajectory has to change is shown changed, with a warning: MPRAGE's shot length to a whole divisor of matrix kz, its acceleration to what the elliptical shutter allows, GRAPPA 2D's acceleration to a whole number.

Step 3: Looking at k-space

The **k-space** tab shows the samples: black where nothing is sampled yet, in color where it is.

- **orientation**: the plane shown, 1 ky-kz (the phase encodings), 2 kx-ky and 3 kx-kz (each readout a whole kx line). A 3D trajectory opens in plane 1, a 2D one in plane 2.
- **color**: **order** (when each point is first sampled: blue first, red last), **count** (how often: green once, toward red the most), or **shot** (the shot that samples it: MPRAGE). The color bar beside the image gives the scale.
- **Play** fills k-space sample by sample, in the order acquired; press it again (**Pause**) to stop. **speed** runs the whole trajectory in 20 s (left) to 1 s (right). **sample** (the field and the slider) shows k-space as it is after that many samples. The mouse wheel over a slider moves that slider, one step per notch.
- The box under the animation sums the trajectory up: the kind of LUT file and the PE1 order it is for, the samples, the acceleration (nrLUT) or the different points sampled (exLUT), the ky and kz ranges, and the shots, spokes or spirals.
- **file**: the name the LUT file gets (Step 4).

Step 4: Saving the LUT file

The name under the k-space image is made from the trajectory: short, so that the scanner's list shows it whole.

Trajectory	Example name
GRAPPA 2D, Gaussian 2D	`GRAPPA_256_R2.4`, `GAUSS_256_R2.5` (matrix ky, the acceleration reached)
GRAPPA 3D, CAIPIRINHA 3D, Poisson 3D	`GRAPPA_128x128_R3.6`, `CAIPI_128x128_R3.6`, `POISSON_192x192_R3.0` (matrix ky x kz, the acceleration reached)
MPRAGE 3D	`MPRAGE_192x192_R4.0_S64` (and the shot length)
Radial 3D, Spiral 3D	`RADIAL_64x64`, `SPIRAL_64x64` (matrix ky x kz)

The name can be typed; characters other than letters, digits, _, - and `.` become _, and `.txt` is added. A name of more than 24 characters gives a warning: the scanner's list may not show it whole.

Press **Save LUT**. On the scanner computer the file goes into the scanner's LUT folder, `c:\smis\dev\LUT`; on another computer TRAJapp asks where, every time, unless a folder is set in App settings (section 10). A file of the same name is replaced only after a yes. The message box says where it went, its samples, and the kind of LUT with the PE1 order to set for it. The lamp turns green.

Step 5: Loading a LUT file

Press **Load LUT** and choose a LUT file (`.txt`, nrLUT or exLUT; its numbers one per line, or separated by spaces, commas or semicolons). TRAJapp shows how it fills k-space, and finds the trajectory and parameters that make it: they are set in the Parameters tab, and **Generate** then makes the same file again. A file made with settings the window does not offer is named as the trajectory it looks like, with a warning and the parameters read from it; a file that is none of the trajectories is shown as it is.

8. Using a LUT file on the scanner

A scan acquires its phase encodings in the order of a LUT file when its sequence is set to read one: the FLASH sequence (version 635e or later). Two settings choose it, among the sequence's parameters in Preclinical Scan:

- **PE1 order**: the order of the phase encodings. Of its choices (**Seq**, **Cen**, **ZigZag**, **LUT**, **ExLUT**, **NonRegLUT**), two read TRAJapp's files:
  - **NonRegLUT** for an nrLUT file: fewer entries than Views x Views 3D, the scan undersampled;
  - **ExLUT** for an exLUT file: as many entries as Views x Views 3D, in the trajectory's order (suited to undersampling afterwards and to splitting the scan into dynamics).
- **Lookup Table**: the file, by its name (the files of the scanner's LUT folder, `c:\smis\dev\LUT`).

Trajectory	LUT file	PE1 order	Views	Views 3D
GRAPPA 2D, Gaussian 2D	nrLUT	**NonRegLUT** (5)	matrix ky	– (a 2D scan)
GRAPPA 3D, CAIPIRINHA 3D, Poisson 3D, MPRAGE 3D	nrLUT	**NonRegLUT** (5)	matrix ky	matrix kz
Radial 3D, Spiral 3D	exLUT	**ExLUT** (4)	matrix ky	matrix kz

The number in brackets is the PE1 order as the scan stores it (`pe1_order` in its parameters); TRAJapp's messages and summary give it too. A scan with more than one repetition (dynamics) repeats the file's order each time: the file holds the order of one.

To use a trajectory:

1. Save it with **Save LUT** on the scanner computer, or copy the `.txt` file into `c:\smis\dev\LUT` there.
2. In Preclinical Scan, set the scan's **Views** (and in 3D **Views 3D**) to the trajectory's matrix, as in the table: matrix ky and matrix kz.
3. Set **PE1 order** to **NonRegLUT** or **ExLUT**, as in the table.
4. Choose the file under **Lookup Table**.
5. Run the scan. A NonRegLUT scan acquires only the samples in the file, so its scan time shrinks by about the acceleration; an ExLUT scan acquires as many as a fully sampled scan, in the trajectory's order (a radial or spiral order samples the center more often and leaves some outer points out).

The scanner keeps a copy of the LUT file with the scan's raw data, beside its MRD file, so the scan can be reconstructed later with the trajectory it was made with.

PE1 order LUT reads no file: for 2D undersampling the ky of each line acquired is typed or pasted into the array of **Views** (the **...** button beside it: as many elements as Views, separated by spaces, commas, semicolons or new lines; missing ones become 0). TRAJapp's 2D trajectories are made for NonRegLUT,

with their file under Lookup Table.

9. Reconstruction

P2ROUD reconstructs the scans made with these files: load the scan's MRD file in P2ROUD; it reads the trajectory from the LUT file beside it and reconstructs the undersampled data (for example with compressed sensing). See P2ROUD's manual.

10. The settings

App settings tab:

- **Theme**: **light** or **dark**.
- **Sounds**: the sounds of warnings and errors on or off.
- **LUT folder**: where **Save LUT** saves. On the scanner computer the scanner's LUT folder, `c:\smis\dev\LUT`. **Change** chooses another folder; **Scanner** goes back to the scanner's. Without either (another computer), **Save LUT** asks where, every time.
- **Updates**:
  - **auto update**, ticked as it comes: new versions of TRAJapp arrive by themselves. When one is ready, a window says so with what is new, once TRAJapp is not working (not during **Generate** or **Load LUT**): **Install and Restart** installs it and starts the new TRAJapp; **Later** installs it when you quit TRAJapp, without asking again. Without an internet connection nothing is said and nothing waits. On Windows, TRAJapp installed for all users can be updated only by an administrator; for other users the message box says once where to get the new version to install by hand.
  - **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the TRAJapp menu does the same.
  - **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.

The app's folder (macOS: `~/Library/Application Support/TRAJapp/0.2.9p`; Windows: `%APPDATA%\TRAJapp\0.2.9p`) holds the settings file, `trajappSettings.json`: TRAJapp keeps your choices of the trajectory and the parameters of each trajectory, the view, the theme, the sounds, **auto update** and **notify at start**, the LUT folder and the folders last used for **Load LUT** and **Save LUT**, as the window last had them, for the next start; a new version takes them over, and the message box says so. A settings file that cannot be read is kept beside it under another name (`trajappSettings-unreadable-<date-time>.json`), and the app starts with the standard values and a warning. The folder also holds a log of each session, `trajapp-log-<date-time>.txt`, with every message the message box showed and its time; the newest 100 are kept.

New versions: at its start TRAJapp looks on GitHub for a newer version and says in the message box which version this is and where a newer one is. In the settings file, `checkForUpdates` (`false`) turns this off and `versionCheckTimeout` gives the seconds it may take. Change them in the file while the app is closed.

11. Credits

TRAJapp uses Python, NumPy, SciPy, PySide6 (Qt) and pyqtgraph. Its trajectories are those of Gustav Strijkers' trajectory tools (Python scripts), made into one application. Its window is built as those of the other MRI apps of Amsterdam UMC (Retrospective, P2ROUD, MAPapp, ADCmapp, DSCmapp, SHIMapp).